Optional Lab

Custom-Agent Feature Delivery

30 minutes | Individual or pairs

KavinSchool

Mission

Create three project-scoped Claude Code agents to deliver a notification-preference user story.

TOOLS

Claude Code, Git, Python 3, and `unittest`.

WORKFLOW

Plan, approve, implement, review, and verify.

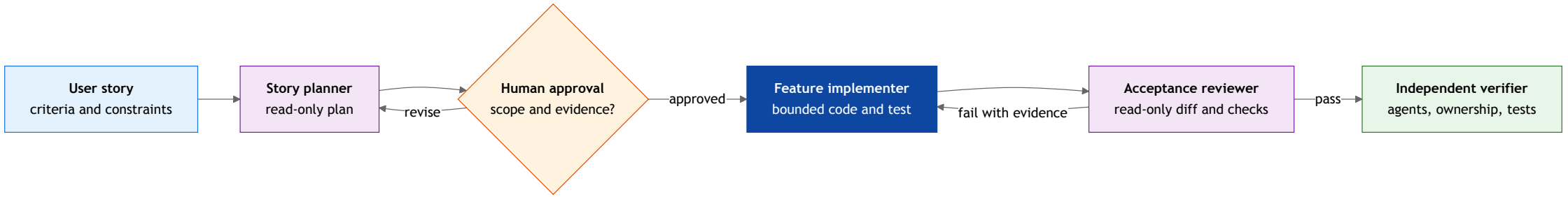
OUTPUT

Agent definitions, a bounded production diff, and passing evidence.

SOURCE

`day1/extensions/custom-agent-feature-delivery/`

Evidence-Gated Agent Chain



Create the Workspace

From the optional lab directory:

```
python3 start/setup_lab.py agent-lab-workspace
cd agent-lab-workspace
python3 -m unittest tests.test_renderers
python3 -m unittest tests.test_preferences
claude
```

Expect the renderer baseline to pass and four preference tests to fail.

30-Minute Run of Show

Time	Checkpoint
0-4	Create the workspace and read the failures.
4-10	Create and inspect three project agents.
10-14	Invoke the planner and approve its boundary.
14-22	Invoke the implementer and inspect its diff.
22-27	Invoke the reviewer and inspect its evidence.
27-30	Run the independent gate and share a decision.

Create Three Project Agents

Use the complete creation prompt in the README. Store definitions under `.claude/agents/`.

Agent	Model	Tools	Responsibility
`story-planner`	Haiku	Read, Glob, Grep	Bound the story and checks
`feature-implementer`	Sonnet	Read, Glob, Grep, Edit, Bash	Change owned code and test it
`acceptance-reviewer`	Sonnet	Read, Glob, Grep, Bash	Review without editing

Reject a planner or reviewer with editing access.

Approve the Plan Before Coding

Ask the main session to delegate `story.md` analysis to `story-planner` and stop after returning the plan.

Approve only when the plan:

1. assigns only `preferences.py` and `router.py`;
2. treats tests as immutable;
3. covers every preference case;
4. names focused and full verification commands.

Revise weak assumptions before invoking the implementer.

Implement, Then Review Independently

IMPLEMENT

Invoke `feature-implementer` on the approved plan; inspect the real diff and focused result.

REVIEW

Have `acceptance-reviewer` inspect `git diff` from `lab-base`, run focused and full checks, and grade every criterion.

REPAIR

Return exact failure evidence without editing; route the smallest correction to the implementer.

Verify and Share Evidence

```
python3 verify_lab.py
git diff --name-only lab-base
git ls-files --others --exclude-standard
```

Expected marker: `CUSTOM AGENT LAB VALID`

Show the three definitions, five allowed changed files, and verifier output. Explain one assumption you rejected, revised, or verified independently.

Definition of Done

If blocked, inspect the failing test, agent definition, delegation evidence, diff boundary, repair prompt, then `solution/`.

AGENTS

Three valid definitions use role-appropriate models and tools; the transcript shows three delegations.

APPROVAL

Human approval occurs before implementation.

SCOPE

Only agent definitions, `preferences.py`, and `router.py` change.

EVIDENCE

Focused, full, and independent checks pass.

Return for the Debrief

Bring the agent chain, the bounded diff, and one evidence-based decision.

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